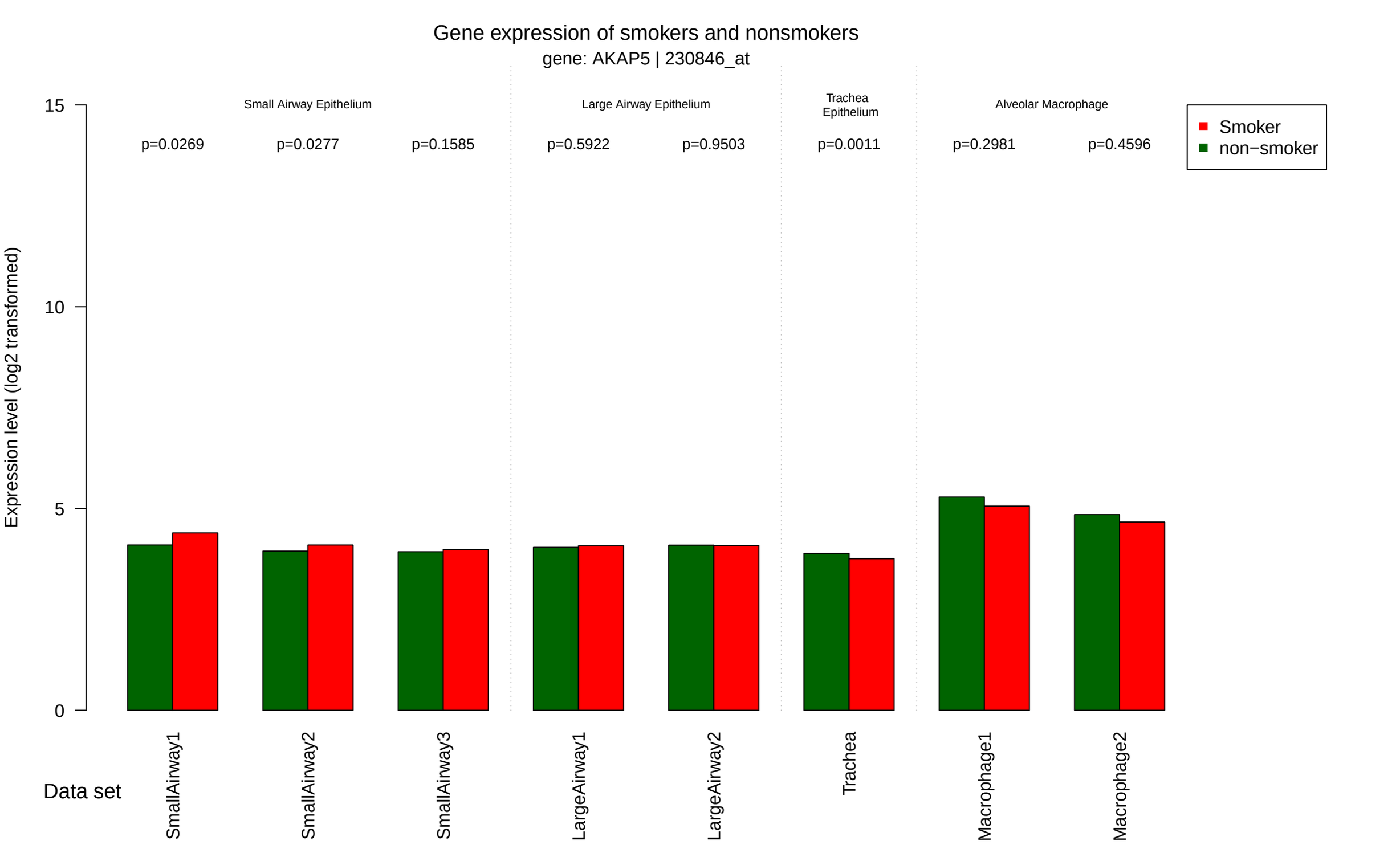
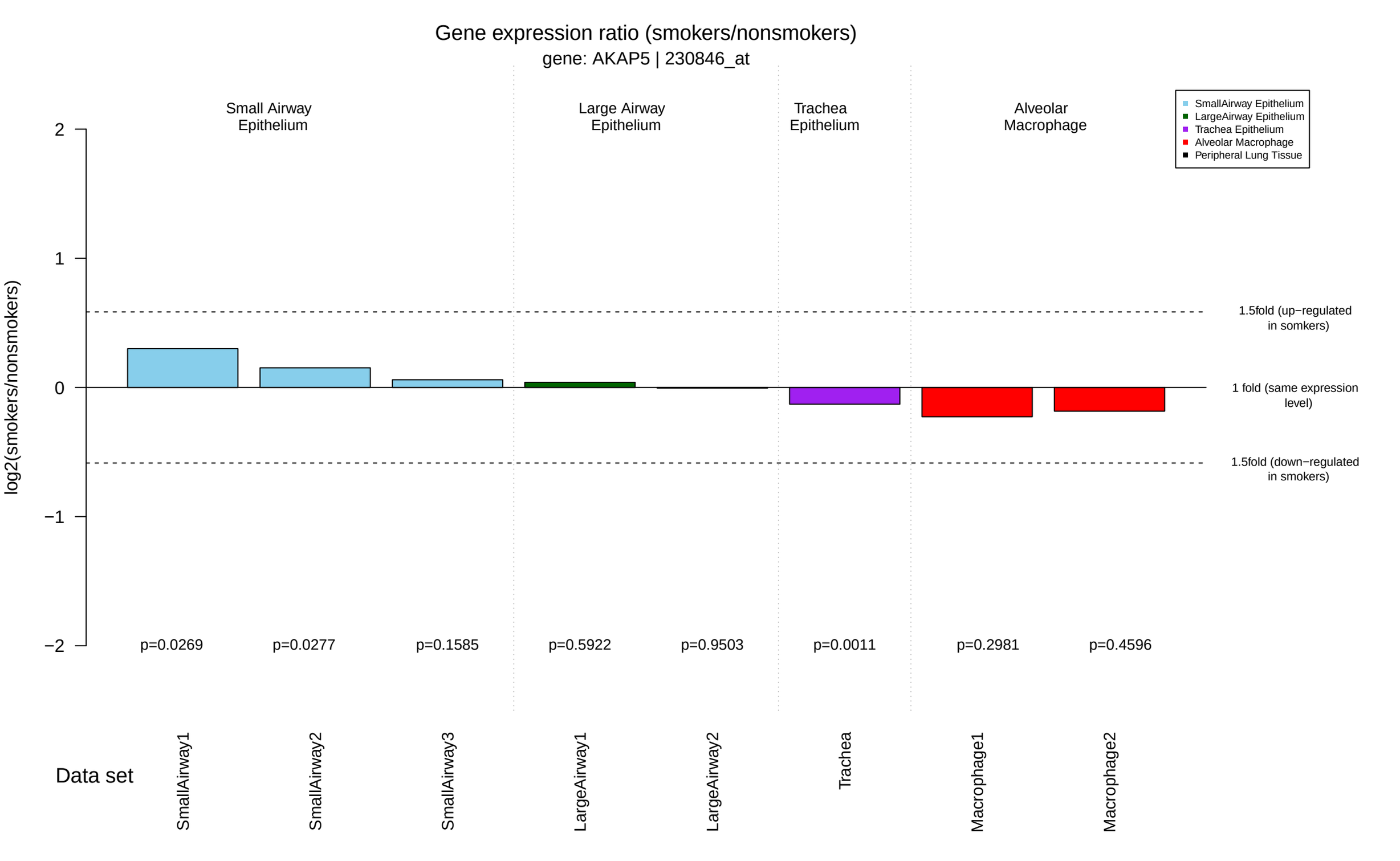
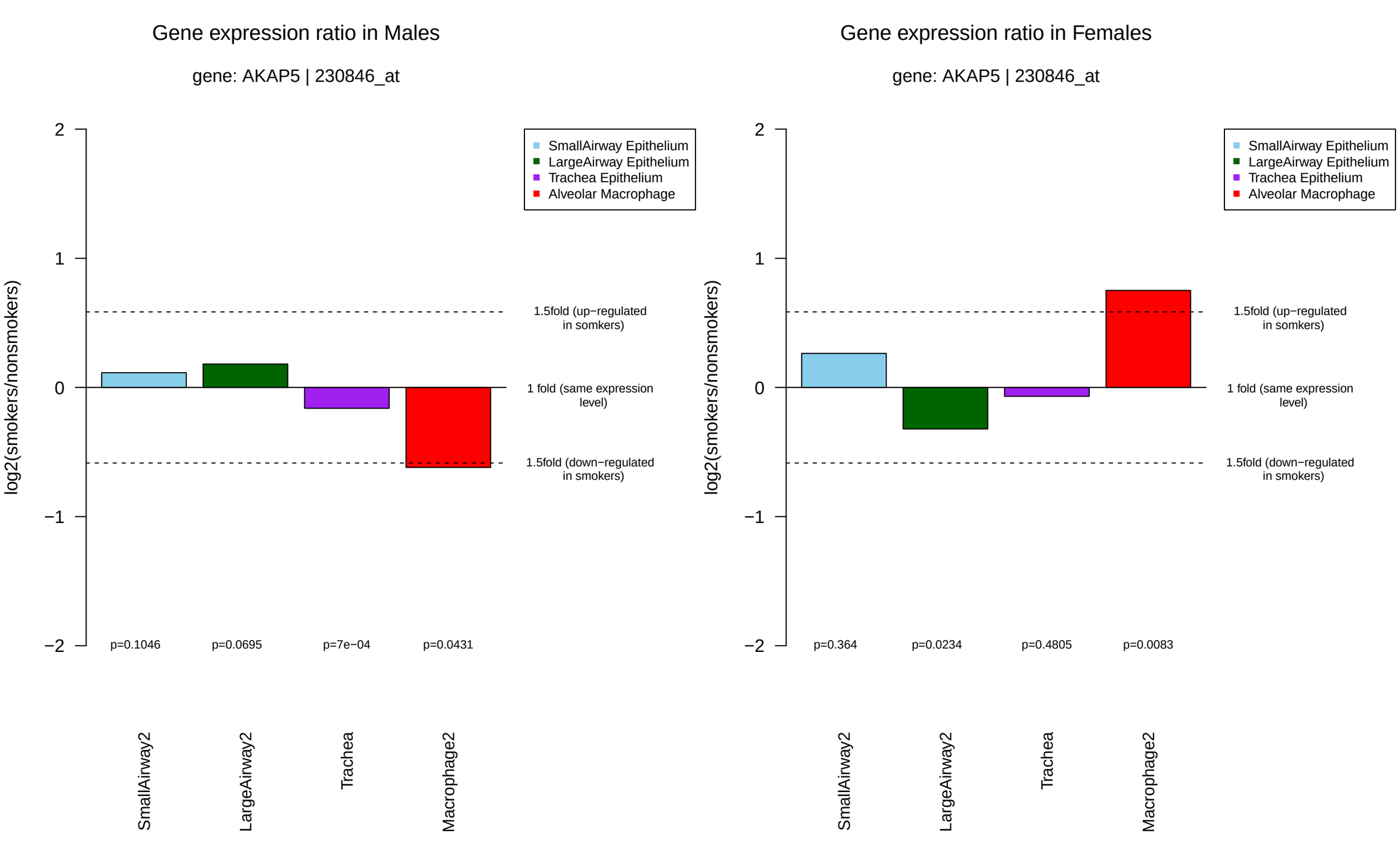
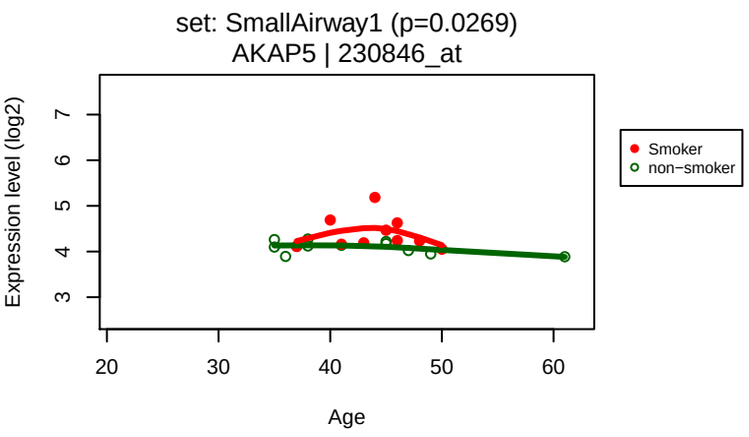
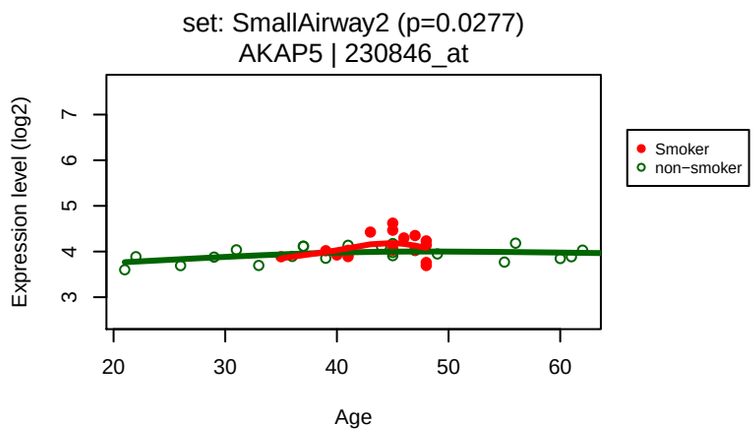
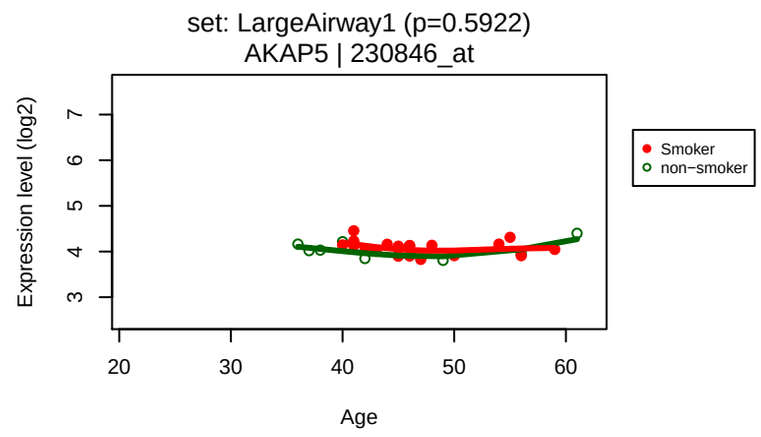

Gene expression of smokers and nonsmokers
gene: AKAP5 | 230846_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0269
p=0.0277
p=0.1585
p=0.5922
p=0.9503
p=0.0011
p=0.2981
p=0.4596
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

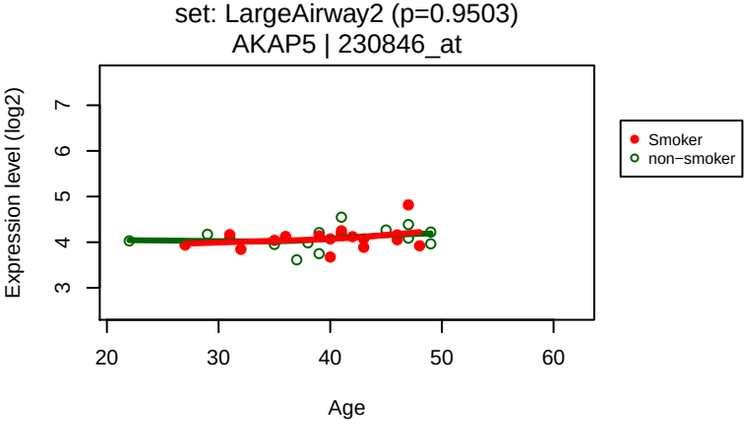

Gene expression ratio (smokers/nonsmokers)
gene: AKAP5 | 230846_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0269
p=0.0277
p=0.1585
p=0.5922
p=0.9503
p=0.0011
p=0.2981
p=0.4596
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

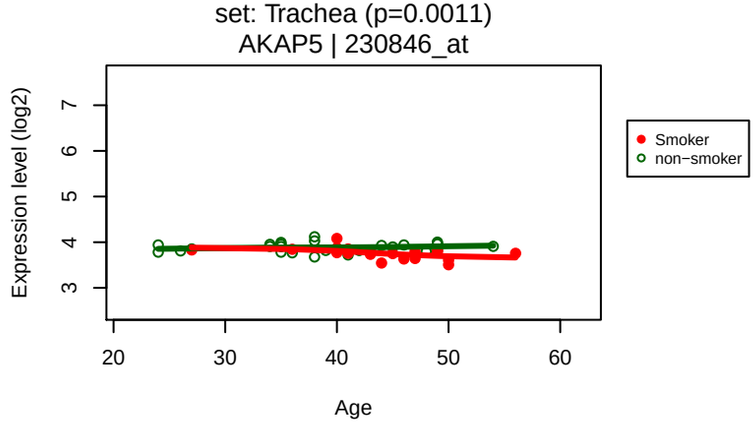

Gene expression ratio in Males
gene: AKAP5 | 230846_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1046
p=0.0695
p=7e−04
p=0.0431
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP5 | 230846_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.364
p=0.0234
p=0.4805
p=0.0083
SmallAirway2
LargeAirway2
Trachea
Macrophage2

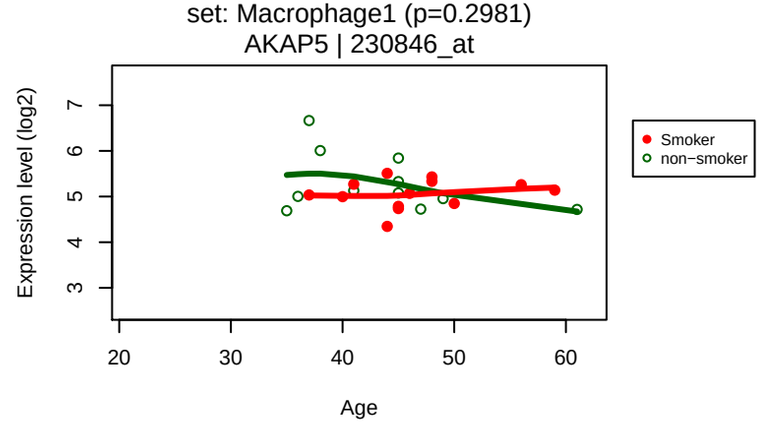

set: SmallAirway1 (p=0.0269)
AKAP5 | 230846_at
Expression level (log2)
Age
Smoker
non−smoker

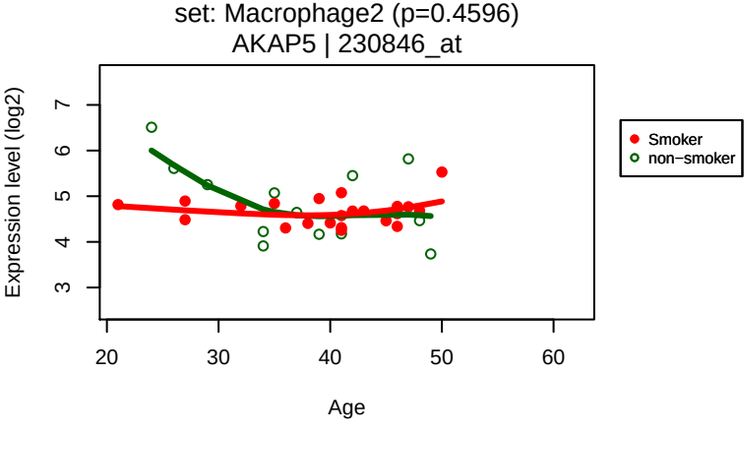

set: SmallAirway2 (p=0.0277)
AKAP5 | 230846_at
Expression level (log2)
Age
Smoker
non−smoker

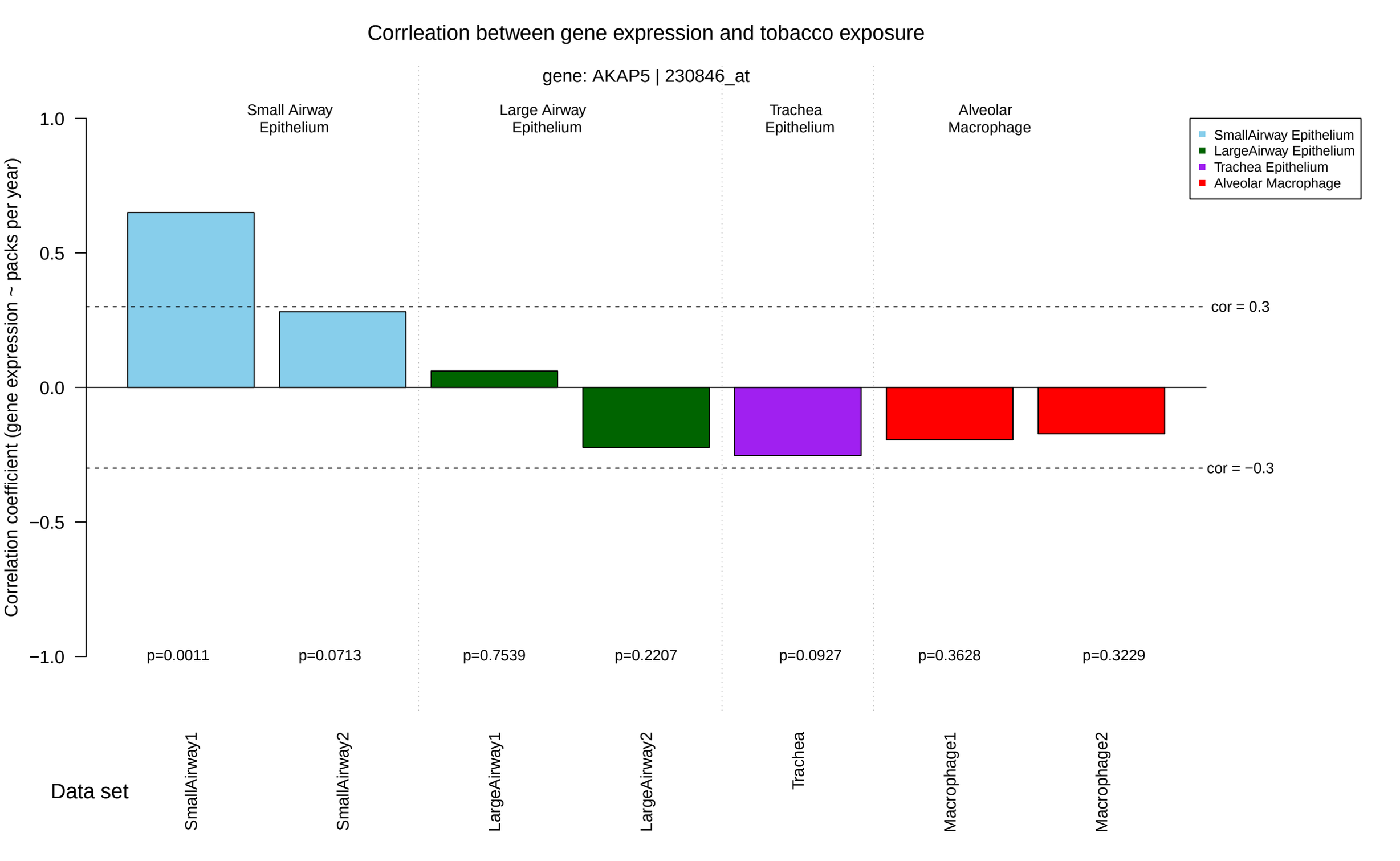

set: LargeAirway1 (p=0.5922)
AKAP5 | 230846_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9503)
AKAP5 | 230846_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0011)
AKAP5 | 230846_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2981)
AKAP5 | 230846_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4596)
AKAP5 | 230846_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP5 | 230846_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0011
p=0.0713
p=0.7539
p=0.2207
p=0.0927
p=0.3628
p=0.3229
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2